Progressive Learning Vision Transformer for Open Set Recognition of Fine-Grained Objects in Remote Sensing Images

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***Abstract*— Open set recognition (OSR) aims to classify known classes and recognize unknown classes simultaneously. Existing OSR methods have primarily focused on learning decision boundaries based on overall feature representations, and have achieved good performance on various coarse-grained image datasets. However, the overall feature representations of objects in fine-grained image datasets are highly similar, making it difficult to distinguish between known and unknown classes by overall feature-based decision boundaries. To address this problem, we propose a progressive learning vision transformer (PLViT) with a coarse-to-fine optimization strategy. In PLViT, the overall feature representations are first optimized in the distance space to learn the initial decision boundaries. Then, a context-aware patch selection module is designed to locate the discriminative part regions. Afterward, the multilayer representations of each selected patch are aggregated according to the self-attention weights, and input into the last transformer layer to extract local feature representations. Finally, overall and local feature representations are adaptively fused and optimized in the angular space to further refine the decision boundaries. Experimental results on four fine-grained remote sensing object recognition datasets show that PLViT outperforms state-of-the-art methods.**

***Index Terms*— Fine-grained visual classification (FGVC), open set recognition (OSR), progressive learning vision transformer (PLViT), remote sensing.**

I. Introduction

DEEP learning methods have recently made continuous progress in object recognition tasks [1], [2], [3]. Conventional methods assume that the training data can cover all classes of the test samples, i.e., the closed set setting. However, in real-world applications, the occurrence of unknown class samples beyond the known training classes is unavoidable. Therefore, the unknown class samples will be misclassified into one of the known classes, causing a serious degradation of reliability. Open set recognition (OSR) [4] has been proposed to solve this pressing problem, which aims not only to correctly classify the known classes but also to accurately recognize the unknown classes.

Manuscript received 20 June 2023; accepted 23 August 2023. Date of publication 28 August 2023; date of current version 8 September 2023. This work was supported in part by the National Natural Science Foundation of China under Grant U20B2067, in part by the Innovation Foundation for Doctor Dissertation of Northwestern Polytechnical University under Grant CX2023015, and in part by the Cultivation Foundation for Excellent Doctoral Dissertation of the School of Automation of Northwestern Polytechnical University. *(Corresponding author: Zhunga Liu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3309091

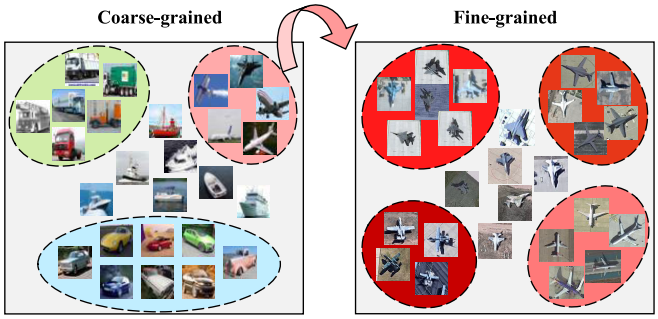

Fig. 1. Difference between the OSR on coarse-grained and fine-grained datasets. Unlike the OSR on coarse-grained datasets, known (e.g., F-16 fighter, B-1 bomber) and unknown classes (F-22 fighter) on fine-grained datasets belong to the same coarse-grained class (aircraft), and are highly similar in overall feature representations.

Existing deep learning-based OSR methods can be divided into two types [5]: the discriminative methods and the generative methods. Discriminative methods focus on optimizing the representations of known classes [6], [7], [8], while generative methods focus on reconstructing/estimating the representations of known/unknown classes [9], [10], [11]. Both types of methods can effectively learn the decision boundaries for OSR based on the overall feature representations, and their performances have been evaluated on various coarse-grained image datasets [12], [13], [14] by setting different basic-level categories with noticeable visual differences (e.g., ship, car, or airplane) as known and unknown classes. As shown in Fig. 1, for OSR on coarse-grained datasets, the interclass differences in overall feature representations are significant, making it easier to recognize unknown classes when classifying known classes. Compared to such unawareness of some basic-level categories, OSR on fine-grained datasets is a more realistic scenario. For OSR on fine-grained datasets, the known and unknown objects, referred to as fine-grained objects, belong to multiple subordinate categories within the same basic-level category (e.g., different types of ships or different models of cars). In this scenario, the objects of known and unknown classes are highly similar in overall feature representations and only differ in subtle local details. Therefore, the involvement of local discriminative features is essential for the learning of decision boundaries for OSR on fine-grained datasets, which is neglected by existing methods.

The extraction of local discriminative features is regarded as the key to fine-grained visual classification (FGVC). As summarized in [15], existing FGVC methods are mainly

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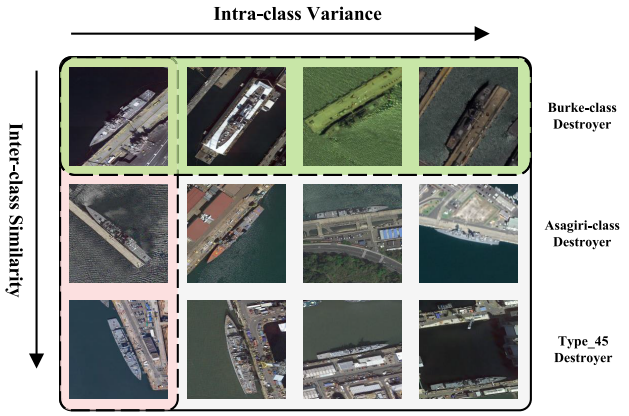

Fig. 2. Key challenges for OSR of fine-grained object in remote sensing images. Each row corresponds to one class in the FGSCR-42 dataset for fine-grained ship recognition.

implemented through feature-encoding or localization-classification subnetworks. With the utilization of localization annotations or weakly supervised learning methods, the localization-classification subnetworks-based methods are more interpretable than the feature-encoding-based methods. However, the manual annotation of localization information is labor-intensive. As a result, current studies have focused more on exploring weakly supervised methods to extract local discriminative features with only image-level labels. Recently, vision transformer (ViT) [3] and its extensions have achieved advanced performance in various computer vision tasks such as image classification [16], object detection [17], and semantic segmentation [18]. The transformer encoder, which is composed of multiple layers of multihead self-attention (MSA) and multilayer perceptron (MLP) blocks, can directly capture the importance of each image patch in regards to the category information, thus implicitly localizing the regions for local discriminative feature extraction. He et al. [19] first leveraged this innate advantage of ViT-based framework for FGVC, but their exploration is limited to the closed set setting.

The recognition of fine-grained objects in remote-sensing images is of great value in both military and civilian applications. Especially in military applications, the continual iterations of novel classes always make the existing training dataset "outdated," so the implementation of OSR becomes particularly necessary. Among diverse remote sensing data sources, optical and synthetic aperture radar (SAR) images are the two most widely used sensory data. Optical remote sensing images contain rich spatial texture information with higher resolution, while SAR images are available in all-weather conditions both day and night. As demonstrated in Fig. 2, besides the high interclass similarity that commonly exists in fine-grained datasets, the changes of resolution, view, and background in remote sensing images also lead to large intraclass variability, making the OSR of fine-grained objects in remote sensing images more challenging.

In this article, we propose a progressive learning ViT (PLViT), which is trained with a coarse-to-fine optimization strategy following the human perception mode. Different from existing OSR methods, PLViT leverages both the overall and local feature representations to learn the decision boundaries for OSR of fine-grained objects. Initially, compact and well-clustered representations are formed by matching the overall features with their corresponding memories in the distance space. This enables the learning of initial decision boundaries and the capture of the category-related importance of each patch. Subsequently, the discriminative part regions are located by the context-aware patch selection (CAPS) module, which integrates MSA weights across both layers and patches. Furthermore, the multilayer representations of the selected patches are aggregated to provide more comprehensive descriptions for local feature extraction. Finally, the overall and local features can be adaptively fused by learning the sufficiency of overall features. The optimization of fusion features is carried out in the angular space to avoid scale inconsistency with overall-based memories in the distance space. Based on the fusion features with sufficient intraclass compactness and interclass separability, the discriminability of the decision boundaries can be effectively improved.

Our main contributions can be summarized as follows.

1) We propose a PLViT that leverages both the overall and local features for OSR of fine-grained objects in remote sensing images. To the best of our knowledge, this is the first exploration for OSR of fine-grained objects in remote sensing images, which effectively meets the growing need for better reliability when deploying deep learning methods in real-world applications.
2) We propose a context-aware patch selection module to accurately locate discriminative part regions by MSA weights across both layers and patches. The aggregation of multilayer representations can provide more comprehensive descriptions of selected patches than single-layer representations, which effectively promotes the extraction of local features for fine-grained objects.
3) We propose a coarse-to-fine optimization strategy to train the network following human perception mode. The processes of overall feature representation and local feature extraction are collaboratively associated through MSA weights in a progressive manner, and the scale inconsistency between overall and local features is effectively avoided, leading to more discriminative decision boundaries than end-to-end training.

We validate the proposed PLViT on four fine-grained object recognition datasets in SAR and optical remote sensing images: Moving and stationary target recognition (MSTAR) [20], multi-type aircraft remote sensing images (MTARSI) [21], fine-grained ship classification (FGSC-23) [22], and FGSC in remote sensing images (FGSCR-42) [23]. The experiment results show that PLViT can achieve state-of-the-art performances with both closed-set and open-set settings.

The rest of this article is organized as follows. We give a brief introduction of the related works in Section II. Then, we describe the technical details of the proposed method in Section III and report the experiment results in Section IV. Finally, we conclude this article in Section V.

II. Related Works

In this section, we provide a brief review of the related works of our proposed method, including OSR, FGVC, and recognition of objects in remote sensing images.

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A. Open Set Recognition

In addition to the earlier proposed machine learning-based OSR methods [4], [24], [25], [26], current OSR methods are mainly based on convolutional neural network (CNN) backbones and can be divided into two types: discriminative methods and generative methods.

1) Discriminative Methods: Bendale and Boult [6] proposed OpenMax as the first deep learning-based solution to OSR. They fitted a Weibull distribution based on the distance of the activation vectors to estimate the unknown probability. Shu et al. [27] proposed a 1-vs-rest final sigmoid layer to reduce the open space risk. Yang et al. [7] proposed generalized convolutional prototype learning (GCPL) to improve the intraclass compactness of feature representations for OSR. In contrast to GCPL, Chen et al. [8] proposed a novel learning framework named reciprocal point learning (RPL) to model the extra-class space, and extended it with adversarial margin constraint [28] for better OSR performance. Miller et al. [29] proposed class anchor clustering (CAC) in the logit space to improve the performance of distance-based OSR classifiers. Liu et al. [30] proposed orientational distribution learning (ODL) to optimize the spatial distribution of feature representations for OSR.

2) Generative Methods: Different from discriminative methods, generative methods learn the decision boundaries by reconstructing/generating the representations of known/unknown classes. Ge et al. [9] proposed G-OpenMax, which generated the training samples of unknown classes with the generative adversarial network (GAN) [31], but its improvement is limited when applied to natural images. Neal et al. [10] proposed an encoder–decoder architecture to generate the counterfactual images as the synthetic training data of unknown classes. Yoshihashi et al. [11] utilized the latent representations for reconstruction and realized the recognition of unknown classes while maintaining the classification accuracy of known classes. However, the subtle local details of fine-grained objects are hard to estimate without the prior description of unknown classes during training.

Existing OSR methods are not suitable for OSR of fine-grained objects as their decision boundaries are learned based on overall feature representations. Some recent attempts have been made based on class activation mapping value [32] and object detection [33] to improve the OSR performance on fine-grained datasets. However, there is a void in how local feature representations can be effectively extracted and used to learn decision boundaries for OSR of fine-grained objects.

B. Fine-Grained Visual Classification

Existing FGVC methods mainly capture the subtle differences in two ways: end-to-end feature encoding and localization-classification subnetworks.

1) End-to-End Feature Encoding: The methods based on end-to-end feature encoding aim to capture the subtle differences by learning more discriminative feature representations. Lin et al. [34] proposed bilinear CNN (B-CNN), which utilized the outer product of the outputs of two feature extractors to capture the pairwise interactions in a high-dimensional feature space. However, obtaining high-dimensional features by outer product will greatly increase the number of parameters, so its universality to real applications is limited. To overcome this limitation, Gao et al. [35] used Tensor Sketch projection to compact the bilinear representation through end-to-end back-propagation. Kong and Fowlkes [36] proposed a low-rank bilinear pooling model to effectively reduce the computational complexity. Cui et al. [37] proposed a general kernel pooling method by compact explicit feature mapping approaches. Yu et al. [38] proposed a cross-layer bilinear pooling to mutually reinforce the interlayer feature interaction and representation learning. However, these methods lack interpretability as there is no direct link between the enriched feature representations and the subtle differences.

2) Localization-Classification Subnetworks: These methods aim to locate the discriminative part regions by a localization subnetwork and then extract the local feature representations. In collaboration with a classification subnetwork, the recognition performance can be improved with better interpretability. Early works [39], [40], [41] used additional annotations, e.g., bounding boxes and segmentation masks, to train the localization subnetwork in a fully supervised way. Whereas, obtaining manual annotations is time-consuming and labor-intensive. In this case, recent studies focused on taking advantage of weakly supervised learning to locate discriminative part regions with only image-level labels. Jaderberg et al. [42] proposed a spatial transformer network (STN) to select the regions that are most relevant to the classification result. Fu et al. [43] proposed a recurrent attention CNN (RA-CNN) to learn the accurate detection and effective representation of discriminative part regions in a mutually reinforced way. Zheng et al. [44] proposed a multiattention CNN (MA-CNN) that jointly learned part proposals by channel grouping subnetwork and feature representations. He et al. [19] proposed a transformer-based architecture to naturally locate discriminative regions for fine-grained objects from overlapping image patches.

Although the recognition performance has been improved by these methods, the importance of different part regions to the classification results cannot be measured directly by normal CNN backbones, and training the specially designed localization subnetwork will bring extra parameters. Besides, preserving local neighboring structures by overlapping image patches leads to additional computational burdens. In contrast, our proposed PLViT is more intuitive. The relationships between different patches and their importance to the classification result can be effectively measured through layers of MSA blocks without increases in parameter numbers and computational burdens.

C. Recognition of Objects in Remote Sensing Images

The recognition of objects in remote sensing images has gained increasing interest over the past few years due to its great value in both military and civilian areas. Among existing studies, the recognition of objects in SAR and optical remote sensing images are two of the most typical scenarios.

1) Recognition of Objects in SAR Images: Traditional SAR automatic target recognition (ATR) methods are inefficient

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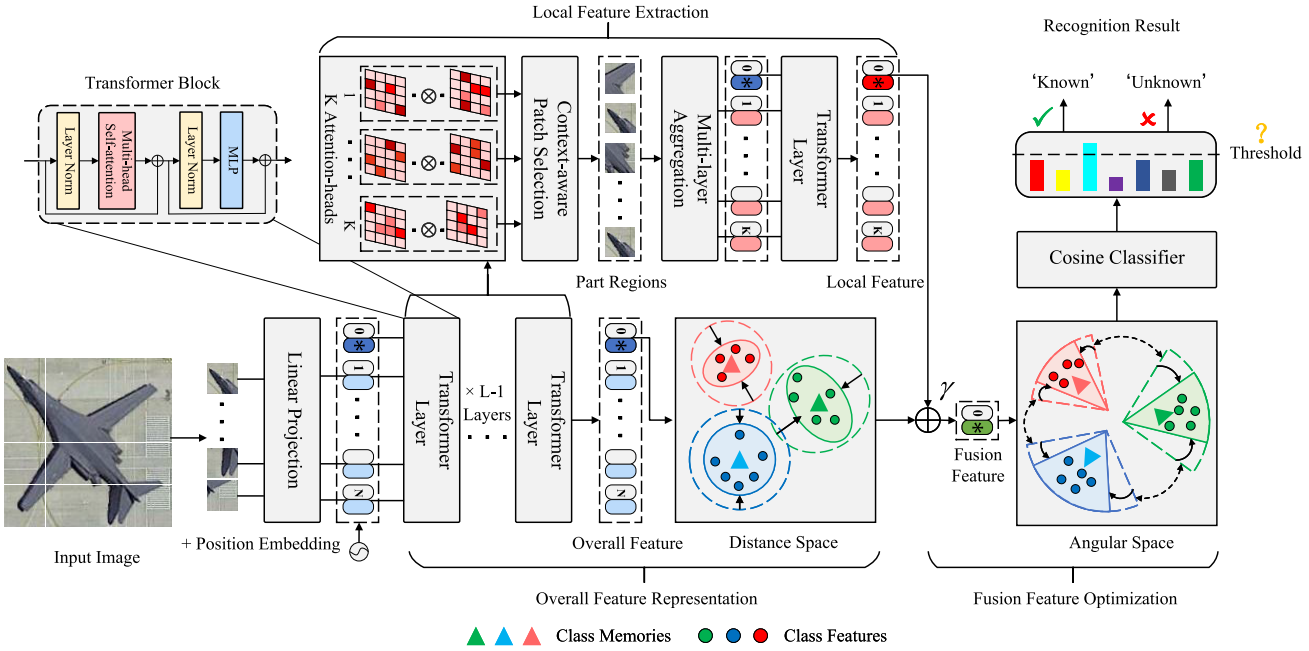

Fig. 3. Overall framework of our proposed PLViT. In a human-like way, the recognition process of PLViT is implemented from coarse to fine through three main parts: overall feature representation, local feature extraction, and fusion feature optimization.

and cannot be optimized end-to-end. To address this problem, recent studies adopted deep learning methods for SAR ATR and achieved significant improvements. Chen et al. [45] proposed A-ConvNets to automatically learn hierarchical features for SAR target recognition. Pei et al. [46] proposed a multiview framework to train the network limited data. Liu et al. [47] proposed a fusion learning framework to take full advantage of physical knowledge and deep features for SAR ATR. Some initial attempts have also been carried out to extend SAR ATR into the open world. Dang et al. [48] proposed an open-set incremental learning method to detect and learn new categories continually. Geng et al. [49] proposed Generative models via Random Sampling Combination (GvRSC) to estimate the distribution of unknown objects in SAR images.

2) Recognition of Objects in Optical Images: The recognition of objects in optical images has also attracted great research interest due to the easy-to-access and high-resolution properties of optical images. Zhang et al. [22] proposed the FGSC-23 dataset which contains 23 ship categories with an attribute-guided multilevel enhanced feature representation network (AMEFRN) to increase interclass differentiation. Di et al. [23] investigated a comprehensive number of ships in remote sensing images and established the FGSCR-42 dataset with a wider variety of ship categories. Hong et al. [50] proposed a fusion framework to combine representations from different perspectives, which broke through the performance bottleneck of single CNN models. Li and Bian [51] proposed a foreground-aware feature map reconstruction network (FRN) for few-shot FGSC-23. Based on the two datasets mentioned above, Xiong et al. [52] proposed an explainable attention network with a causal multihead attention model (CMAM) for fine-grained ship recognition in remote sensing images, which can provide an intuitive understanding of the decision-making process. In addition to the recognition of ships, a new dataset for aircraft recognition was established [21], with challenges like complex backgrounds and different spatial resolutions.

Besides single-modal scenarios, the recognition performance of remote sensing images can be further improved by multimodal learning frameworks [53] in complex scenes. Although some studies in open-world scenarios have been proposed, their experimental setups, which typically involve a single dataset or a limited number of unknown classes, are relatively simple. In contrast, our exploration of multiple datasets with various setups is more comprehensive, and the visualization results can provide a clear explanation of the recognition process.

III. Progressive Learning Vision Transformer

The architecture overview of our proposed PLViT is shown in Fig. 3. We leverage both overall and local features to learn decision boundaries for OSR of fine-grained objects. The overall training process can be divided into two stages. First, patch embeddings are input to the transformer encoder to learn overall feature representations in the distance space, which corresponds to the coarse optimization stage. Then, the model is trained to extract local features from discriminative part regions and optimize adaptive fusion features in the angular space during the fine optimization stage. In Sections III-A–III-B, we first review how the ViT model works for image classification tasks. Then, each crucial part of our proposed method is introduced in detail.

A. Vision Transformer for Image Classification

The standard transformer encoder in natural language processing (NLP) requires the input to be the 1-D sequence of token embeddings. First, in order to apply this structure to image classification tasks, the input image $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ is divided into different patches and reshaped as a sequence

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$\mathbf{x}_p \in \mathbb{R}^{N_P \times (P^2 \cdot C)}$, where $H \times W$ is the resolution of the image, C is the number of channels, and $P \times P$ is the resolution of the patch. The number of patches is $N_P = HW/P^2$, which also refers to the input sequence length for the transformer encoder. Next, the patches are flattened and mapped to D-dimensional embeddings by a trainable linear projection layer $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}$. Then, an extra learnable class token $\mathbf{x}_{\text{class}} \in \mathbb{R}^D$ is concatenated with the patch embeddings, and the positional information is retained by the position embeddings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N_P+1) \times D}$

$$\mathbf{z}_0 = \left[\mathbf{x}_{\text{class}}, \mathbf{x}_p^1\mathbf{E}, \mathbf{x}_p^2\mathbf{E}, \ldots, \mathbf{x}_p^{N_P}\mathbf{E}\right] + \mathbf{E}_{\text{pos}}. \tag{1}$$

Afterward, the embeddings are input to the transformer encoder containing alternating L layers of MSA and MLP blocks. The output of the lth layer is calculated as follows:

$$\mathbf{z}_l' = \text{MSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1}, \quad l \in \{1, \ldots, L\} \tag{2}$$

$$\mathbf{z}_l = \text{MLP}\left(\text{LN}\left(\mathbf{z}_l'\right)\right) + \mathbf{z}_l', \quad l \in \{1, \ldots, L\} \tag{3}$$

where $\text{LN}(\cdot)$ is the layer normalization operation. Finally, the output state of the class token from the last layer $\mathbf{z}_L^0$ is used as the input to the classification head.

B. Overall Feature Representation

While capturing subtle discriminative differences based on local features is critical for FGVC, the effective representation of overall features is also essential. On the one hand, the high similarity of overall features exists within different subsets consisting of partial fine-grained classes rather than all of them, i.e., some fine-grained classes can be well distinguished from each other based on overall features. On the other hand, for fine-grained classes with high overall similarity, the effective overall feature representation can potentially facilitate the extraction of local discriminative features.

To this end, we use the state of the class token before the last transformer layer $\mathbf{z}_{L-1}^0 \in \mathbb{R}^D$ as the overall feature of the input image x and optimize it in the distance space. Following [7], we initialize a set of learnable parameters $M = \{m_c\}_{c=1}^N \in \mathbb{R}^{N \times D}$ to denote the memories of N known classes. To match the overall features of each class with the corresponding memories, we minimize the Euclidean distance between them by the distance loss L_D

$$L_D\left(\mathbf{z}_{L-1}^0, m_y\right) = \|\mathbf{z}_{L-1}^0 - m_y\|_2^2 \tag{4}$$

where y is the class label of x. We also use the cross-entropy loss L_{CE} to build a direct link between overall features and category information

$$L_{\text{CE}}\left(\mathbf{z}_{L-1}^0, y\right) = y \log\left(\ell\left(\mathbf{z}_{L-1}^0\right)\right) + (1-y)\log\left(1 - \ell\left(\mathbf{z}_{L-1}^0\right)\right) \tag{5}$$

where $\ell(\cdot)$ is the linear classifier. The overall features of the objects can be effectively represented with the joint supervision of L_D and L_{CE}

$$L_{\text{overall}} = L_{\text{CE}} + \lambda_{\text{Dis}} L_D \tag{6}$$

where λ_{Dis} is a weight parameter for the distance loss.

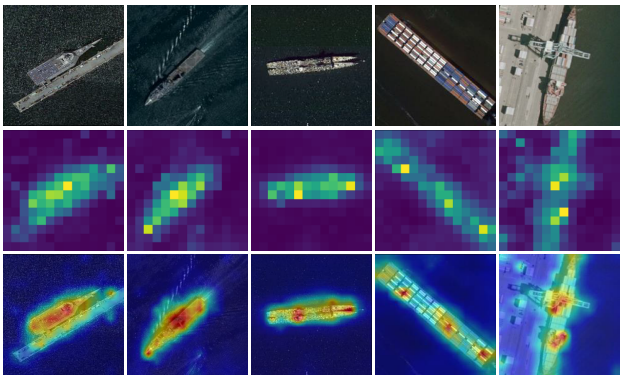

Fig. 4. Visualization of the image samples with the corresponding attention masks and attention maps (best viewed in color).

C. Local Feature Extraction

The importance of different patches to the classification result has been implicitly quantified through the overall feature representation process, which is of great value for the localization of discriminative part regions. The previous work [19] ignores the context information for localization and only takes the high-level representations of the selected patches as the input for local feature extraction. As shown in Fig. 4, the attention weights around the discriminative regions of fine-grained objects are gradually increased to the most critical parts, rather than being highlighted in isolation. This indicates the need to take the context information into account for localization. Moreover, the features in low and high layers contain more detail and semantic information, respectively, and the recognition of fine-grained objects can benefit from the aggregation of multiple-layer representations [54], [55].

Therefore, we propose a context-aware part selection module to locate the discriminate part regions in the image and aggregate their multilayer representations to extract local features. Specifically, for MSA blocks in the previous $L-1$ transformer layers with K self-attention heads, their attention weights can be represented as

$$\alpha_l = \left[a_l^1, a_l^2, \ldots, a_l^K\right], \quad l \in \{1, \ldots, L-1\} \tag{7}$$

$$a_l^k = \left[a_l^{k_0}, a_l^{k_1}, \ldots, a_l^{k_{N_P}}\right], \quad k \in \{1, \ldots, K\} \tag{8}$$

where the multiple self-attention heads can be viewed as different visual perception patterns.

First, we multiply the attention weights of all previous layers recursively into $\alpha_{\text{overall}} \in \mathbb{R}^{K \times (N_P+1) \times (N_P+1)}$

$$\alpha_{\text{overall}} = \prod_{l=1}^{L-1} \alpha_l = \left[a_{\text{overall}}^{k_0}, a_{\text{overall}}^{k_1}, \ldots, a_{\text{overall}}^{k_{N_P}}\right]. \tag{9}$$

Next, we calculate the classification relevance r_k^p of each patch with its context information based on the weights $a_{\text{overall}}^{k_0} \in \mathbb{R}^{(N_P+1)}$ queried by the class token of attention head k

$$r_k^p = a_{\text{overall}}^{k_0}(p) + \eta \text{Mean}\left(a_{\text{overall}}^{k_0}(n_p)\right) \tag{10}$$

where n_p represents the adjacent patches of patch p, and η is the coefficient of the context information. Then, we select the patches with the highest classification relevance as the most discriminative region of each attention head, whose indices are represented as $I = [I_1; I_2, \ldots, I_K]$. For the selected patch with

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index I_k, its multilayer attention weights queried by the class token of the corresponding attention head k can be represented as a_k

$$a_k = \left[a_1^{k_0}(I_k), a_2^{k_0}(I_k), \ldots, a_{L-1}^{k_0}(I_k)\right] \tag{11}$$

where $a_k \in \mathbb{R}^{L-1}$. Afterward, we normalize a_k into $\hat{a}_k$ by layerwise Softmax operation to aggregate the multilayer representations as $\mathbf{z}_k \in \mathbb{R}^D$

$$\mathbf{z}_k = \sum_{l=1}^{L-1} \hat{a}_{k_l} \mathbf{z}_l^{I_k}. \tag{12}$$

Finally, we concatenate the overall feature and the aggregated representations into $\mathbf{z}_{\text{local}}$, and input it to the last transformer layer for local feature extraction

$$\mathbf{z}_{\text{local}} = \left[\mathbf{z}_{L-1}^0, \mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_K\right] \tag{13}$$

and the output state of the class token $\mathbf{z}_L^0$ refers to the local features extracted from the discriminative part regions.

D. Fusion Feature Optimization

As discussed above, the effectively represented overall features may be sufficient to distinguish some fine-grained classes, whereas the local features are more important to distinguish those with high overall similarity. We adopt a linear layer W with $\tanh(\cdot)$ activation function to describe the insufficiency of the overall features as $\gamma \in \mathbb{R}^D$

$$\gamma = \tanh\left(W\left(\mathbf{z}_{L-1}^0\right)\right). \tag{14}$$

Therefore, we can adaptively fuse the local features and the overall features as the fusion feature $\mathbf{z}_F$

$$\mathbf{z}_F = \mathbf{z}_{L-1}^0 + \gamma \otimes \mathbf{z}_L^0 \tag{15}$$

where $\otimes$ is the elementwise product operation. When γ is small, the overall feature may be sufficient enough. Otherwise, the association with local features is indispensable.

The scales of overall-based memories and local features are inconsistent due to the differences in both extraction and representation mechanisms. As the distance metric is sensitive to the change of scale [56], it is inappropriate to still optimize the fusion feature in the distance space. To obviate this problem and learn more discriminative decision boundaries, we optimize the fusion feature in the angular space, where the metric is scale-invariant. Specifically, we initialize another set of learnable parameters $M^* = \{m_c^*\}_{c=1}^N \in \mathbb{R}^{N\times D}$ by the averaged fusion features of each class during the first forward propagation, which denote the fusion memories of N known classes. Both features and memories are converted into angular space by L2 normalization and represented as $\bar{\mathbf{z}}_F$ and $\bar{m}^*$

$$\bar{\mathbf{z}}_F = \frac{\mathbf{z}_F}{\|\mathbf{z}_F\|}, \quad \bar{m}_y^* = \frac{m_y^*}{\|m_y^*\|}. \tag{16}$$

To increase intraclass compactness and interclass separability simultaneously, we minimize the angles between the fusion features and their corresponding memories (positive samples) and maximize the angles between memories of other classes (negative samples). The optimization can be formulated as a prototype-based triplet margin loss L_{Ang} in the angular space

$$L_{\text{Ang}} = \max\left(0, \phi\left(\bar{\mathbf{z}}_F, \bar{m}_y^*\right) + \theta - \min_{j\neq y} \phi\left(\bar{\mathbf{z}}_F, \bar{m}_j^*\right)\right) \tag{17}$$

where $\phi(\cdot)$ represents the angle between two vectors, and θ is the angular margin. Finally, the fine-tuning of the decision boundaries can be achieved by combining L_{Ang} and $\bar{L}_{\text{CE}}$

$$L_{\text{fuse}} = \bar{L}_{\text{CE}}\left(\bar{\mathbf{z}}_F, y\right) + \lambda_{\text{Ang}} L_{\text{Ang}} \tag{18}$$

where $\bar{L}_{\text{CE}}$ is the cross-entropy loss with cosine classifier, λ_{Ang} is a weight parameter for the angular loss.

During testing, the recognition result is obtained through a probability threshold δ. If the maximum class prediction surpasses this threshold, the input sample will be classified into the corresponding known class. Otherwise, it will be classified as an unknown class sample

$$\text{Result} = \begin{cases} \arg\max P(\mathbf{x}), & \text{if } \max P(\mathbf{x}) \geq \delta \\ \text{unknown}, & \text{otherwise} \end{cases} \tag{19}$$

where $P(\mathbf{x})$ is the softmax probability of the object from cosine classifier [57]. Compared with classical linear classifiers, the cosine classifier can prompt the model to extract discriminative visual cues and also improve the intraclass variance of the fusion features by closely matching the normalized feature representations with the classification weight vectors, which is of great value in improving the OSR performance.

IV. Experiments

A. Experimental Setup

1) Datasets: Four datasets are used to evaluate our proposed method and some image samples are shown in Fig. 5:

MSTAR [20] is a dataset that consists of ten fine-grained categories of military vehicles in the X-band and HH polarization SAR images with the 0.3-m resolution.

MTARSI [21] is a dataset that contains 9385 images of 20 aircraft categories from remote sensing images, with complex backgrounds, different spatial resolutions, and pose variations.

FGSC-23 [22] is a FGSC-23 dataset. It comprises 4080 ship slices of 23 fine-grained categories. Their image sizes range from 40 × 40 to 800 × 800.

FGSCR-42 [23] is a dataset for fine-grained ship classification from optical remote-sensing images of 42 ship categories. It contains 9320 images whose sizes range from about 50 × 50 to about 1500 × 1500 pixels with various aspect ratios.

2) General Implementation: The train and test splits for each dataset is the same as mentioned in [20], [21], [22], and [23]. The images in the MSTAR dataset and other datasets are resized to 128 × 128 and 224 × 224, respectively. We use ViT-B-16 (pretrained on ImageNet) as the backbone and split images into patches of size 16. The model is trained for 100 epochs (50 epochs each for coarse stage and fine stage) with batch size 32 by stochastic gradient descent (SGD) optimizer, and the learning rate is initialized as 0.01. The hyperparameters λ_{Dis}, η, λ_{Ang}, and θ is set to 0.01, 0.5, 1, and 0.3, respectively. The experiments are conducted in PyTorch framework with Nvidia RTX 3090 GPU support.

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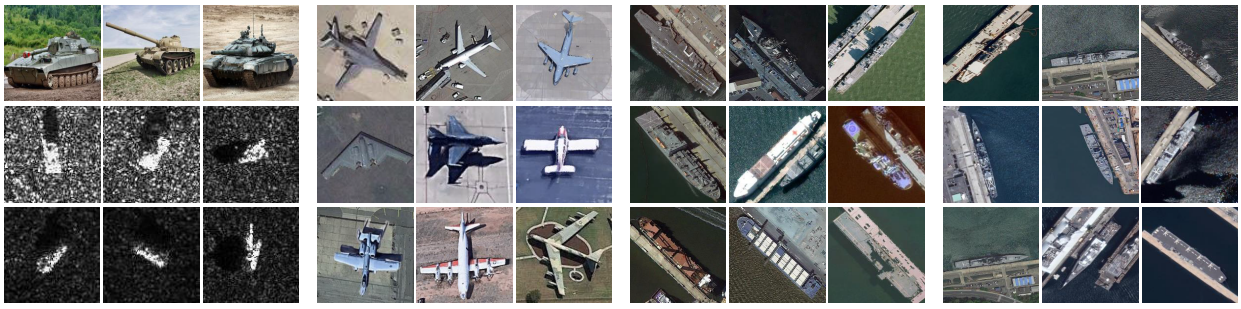

Fig. 5. Image samples of different fine-grained object recognition datasets. The images in the first row of the MSTAR dataset are the optical image of the target, and images in the remaining two rows are SAR images with a depression angle of 15° and 17°. The optical images in the remaining three datasets are richer in texture features with more detailed category information.

B. Experiments With Closed Set Setting

In addition to the evaluation with the open set setting, it is important to ensure the classification accuracy of our proposed method with the closed set setting, which is positively correlated with the OSR performance [69]. So, we report the classification accuracy of the proposed method on these four datasets as normal FGVC tasks, and the experiment results are shown in Table I. The baseline methods include the commonly used CNN architectures and some typical recognition methods for fine-grained objects in remote sensing images and natural images. Compared with the baseline methods, our proposed method can achieve outstanding performance, especially a 9.89% and 4.76% improvement in classification accuracy on MTARSI and FGSCR-42 datasets, respectively. For the MSTAR dataset under standard operating conditions (SOCs) whose classification accuracy is near saturation, our method can still make some improvements. For the FGSC-23 dataset with imbalanced sample distribution between different classes, our method can achieve comparable performance without any specially designed data augmentation method for imbalanced data. These demonstrate the effectiveness of our proposed method for the recognition of fine-grained objects in remote sensing images with the closed set setting.

TABLE I
CLOSED SET CLASSIFICATION ACCURACY ON EACH DATASET

Dataset	Method	ACC
MSTAR	SVM	90.00
	AdaBoost	92.00
	Cond Gauss [58]	97.00
	IGT [59]	95.00
	MSRC [60]	93.60
	MSS [61]	96.60
	MPMC [62]	98.80
	A-ConvNets [45]	99.13
	ViT [3]	98.60
	PLViT	**99.30**
MTARSI	AlexNet	85.61
	GoogLeNet	86.53
	VGG	87.56
	ResNet	89.61
	DenseNet	89.15
	EifficientNet	89.79
	ViT [3]	99.57
	PLViT	**99.68**
FGSC-23	Inception-v3	83.88
	DenseNet121	84.00
	MobileNet	84.24
	Xception [63]	87.76
	FDN [64]	82.30
	ME-CNN [65]	85.58
	LGFFE [66]	89.45
	B-CNN [34]	84.00
	DCL [67]	90.66
	AMEFRN [22]	**91.15**
	ViT [3]	87.41
	PLViT	**91.15**
FGSCR-42	VGG19	77.36
	ResNet-50	84.77
	DenseNet	88.69
	ResNext-50	89.16
	B-CNN [34]	89.53
	RA-CNN [43]	91.63
	DCL [67]	93.03
	TASN [68]	93.51
	ViT [3]	97.47
	PLViT	**98.27**

C. Experiments With Open Set Setting

To prove the effectiveness of our method, we compare PLViT with representative FGVC methods [19], [34], [38], [42], [43] and state-of-the-art OSR methods [6], [7], [8], [28], [29] as follows.

1) *Softmax:* It uses the highest Softmax score with a probability threshold to distinguish between known and unknown classes.
2) *STN [42]:* It learns an affine transformation to select regions that are most relevant to classification.
3) *B-CNN [34]:* It uses a pooled outer product to model part-feature interactions to increase the representation power.
4) *RA-CNN [43]:* It recursively learns the localization and representation of part regions in a mutually reinforced way.
5) *Hierarchical Bilinear Pooling (HBP) [38]:* It integrates cross-layer bilinear features to improve the representation without increasing parameters.
6) *TransFG [19]:* It uses matrix multiplication to self-attention weights for discriminative part selection.
7) *OpenMax [6]:* It fits a Weibull distribution to calibrate the known probabilities and estimate unknown probability.

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TABLE II
COMPARISON OF THE OSR PERFORMANCE OF DIFFERENT METHODS. THE FGVC AND OSR METHODS ARE DIVIDED BY THE DASHED LINE

Methods	Backbone	MSTAR		MTARSI		FGSC-23		FGSCR-42	
		AUROC	OSCR	AUROC	OSCR	AUROC	OSCR	AUROC	OSCR
Softmax	ResNet-50	84.3±2.2	83.6±2.0	86.8±5.5	84.3±4.7	76.8±5.4	69.8±6.2	77.2±3.5	71.4±4.4
STN	ResNet-50	82.8±3.2	82.1±3.5	84.9±4.1	82.3±5.0	71.3±2.9	64.2±4.4	83.2±2.8	80.3±5.1
B-CNN	ResNet-50	77.3±8.2	77.0±8.1	80.7±6.0	79.6±5.9	64.4±4.7	59.9±3.9	79.9±3.1	77.4±2.9
RA-CNN	VGG-19	82.1±4.2	81.9±4.2	85.9±5.8	84.3±6.8	65.8±6.2	57.9±5.6	87.1±2.1	84.3±2.3
HBP	ResNet-50	86.2±3.9	86.0±3.8	89.6±5.3	88.7±5.0	77.5±3.8	73.1±4.6	89.1±1.9	88.3±2.1
OpenMax	ResNet-50	72.1±4.9	76.4±3.5	84.5±3.7	82.9±3.6	68.8±3.3	65.1±3.0	78.0±3.1	72.6±3.8
GCPL	ResNet-50	76.9±7.7	74.6±7.8	80.1±5.6	78.6±6.1	60.5±6.0	44.2±5.4	77.9±7.6	74.3±8.6
RPL	ResNet-50	80.9±6.5	80.5±6.4	78.1±5.6	78.0±5.5	63.4±4.5	45.7±6.0	79.2±3.5	68.9±5.5
CAC	ResNet-50	90.5±2.1	90.1±2.0	89.5±2.9	88.3±2.9	79.3±3.3	76.2±3.6	84.1±2.5	78.8±3.4
ARPL	ResNet-50	87.3±6.2	87.2±6.0	89.2±6.0	88.8±6.7	65.9±2.8	55.5±3.3	79.6±5.7	69.3±7.2
DeiT	DeiT-B	87.8±5.0	87.4±4.9	80.9±2.4	78.6±1.7	63.5±4.1	56.3±4.4	82.6±4.5	75.8±4.1
TransFG	ViT-B-16	89.9±3.1	89.6±3.0	77.4±4.7	77.0±4.9	64.1±5.3	60.2±4.7	86.6±1.3	85.9±1.2
PLViT	ViT-B-16	**92.6±2.7**	**92.4±2.6**	**91.7±3.8**	**91.5±3.7**	**80.6±5.3**	**78.7±5.6**	**94.6±1.0**	**94.2±1.1**

8) *GCPL [7]:* It replaces Softmax with an open-world oriented prototype model to deal with unknowns.
9) *RPL [8]:* It models the extra-class spaces of the known classes and reduce open space risk through multicategory interaction.
10) *ARPL [28]:* It combines Euclidean distance and dot product with adversarial margin constraint to learn more discriminative feature space.
11) *CAC [29]:* It encourages the logit vectors to be tightly clustered around anchored class centers in the logit space.

Following [4], we use openness O^* to describe the ratio of known classes to unknown classes:

$$O^* = 1 - \sqrt{\frac{2 \times |C_{\mathrm{TR}}|}{|C_{\mathrm{TR}}| + |C_{\mathrm{TE}}|}} \tag{20}$$

where C_{TR} represents the number of known classes during training and C_{TE} represents the total number of known and unknown classes during testing. For the MSTAR dataset, the numbers of randomly chosen known and unknown classes are 6 and 4. Besides this, the number of known classes for the other three datasets (MTARSI, FGSC-23, and FGSCR-42) is set to 10. Thus, the openness of each dataset is 13.39%, 18.35%, 22.15%, and 37.98%, respectively.

1) Open Set Recognition: We evaluate the OSR performance by averaging the area under ROC curve (AUROC) and open set classification rate (OSCR) results over five randomized trials.

1) **AUROC** is the most commonly used evaluation metric to evaluate the performance of OSR methods. The effect of all possible probability thresholds is measured comprehensively.
2) **OSCR** indicates the unknown recognition performance with the consideration of the accuracy of known classes based on correct classification rate (CCR) and false positive rate (FPR).

The results are shown in Table II, We can see that PLViT significantly outperforms the methods we compare on each dataset, which indicates that it can accurately classify the known classes and recognize unknown classes simultaneously. In addition, the OSR performance on these four datasets of the compared FGVC methods generally outperforms the compared OSR methods. This proves the importance of the effective extraction and representation abilities for subtle discriminative features. The discriminability of decision boundaries learned based on overall features is limited for the recognition of fine-grained objects in remote sensing images without the complement of local features. Furthermore, although the FGSC-23 dataset contains fewer classes than the FGSCR-42 dataset, the OSR performances of different methods on it are inferior to FGSCR-42. This is mainly because of the existence of category imbalance in FGSC-23 (there are 542 samples of destroyer while only 27 samples of medical ship). When some rare classes with fewer samples are randomly selected as known classes, the learning of decision boundaries will become more difficult, which will increase the confusion between known and unknown classes. Moreover, most methods even degrade the OSR performance of Softmax on MTARSI and FGSC-23 datasets. This indicates that the existing FGVC and OSR methods cannot be directly applied to OSR of fine-grained objects in remote sensing images, due to problems such as the change of resolution/background (MTARSI) and the imbalanced data distribution (FGSC-23).

2) Out-of-Distribution Detection: To further prove the effectiveness of our proposed method, we also evaluate its performance following the out-of-distribution detection (OOD) task [70], which is naturally related to OSR. The results are reported by the following metrics.

1) **True negative rate (TNR) at 95% true positive rate (TPR)** is measured by the value of TNR = TN/(TP + TN) when TPR = TP/(FP+FN) is 95%, where TP, TN, FP, and FN refer to true positives, true negatives, false positives, and false negatives, respectively.

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TABLE III
DISTINGUISHING IN- AND OUT-OF-DISTRIBUTION TEST SET DATA UNDER VARIOUS VALIDATION SETUPS. THE FGVC AND OSR METHODS ARE DIVIDED BY THE DASHED LINE

Methods	Backbone	In: **Civilian** / Out: **Military**					In: **Military** / Out: **Civilian**				
		TNR	AUROC	DTACC	AUIN	AUOUT	TNR	AUROC	DTACC	AUIN	AUOUT
Softmax	ResNet-50	26.7	71.9	69.1	39.9	88.2	39.9	86.9	80.5	93.2	68.4
STN	ResNet-50	16.5	69.6	67.0	37.8	88.7	22.2	78.5	72.6	89.6	71.1
GCPL	ResNet-50	23.8	62.4	61.9	37.7	83.8	11.9	66.1	62.8	86.4	36.2
RPL	ResNet-50	18.8	67.7	63.6	44.1	85.0	31.6	84.0	76.2	93.9	61.4
ARPL	ResNet-50	29.4	80.2	73.6	58.5	90.8	73.7	95.6	89.6	98.4	87.7
ViT	ViT-B-16	49.1	90.3	83.5	78.8	96.0	65.4	93.1	87.7	81.1	96.1
PLViT	ViT-B-16	**76.1**	**95.4**	**88.9**	**91.2**	**98.1**	**94.8**	**98.8**	**95.2**	**99.5**	**96.5**

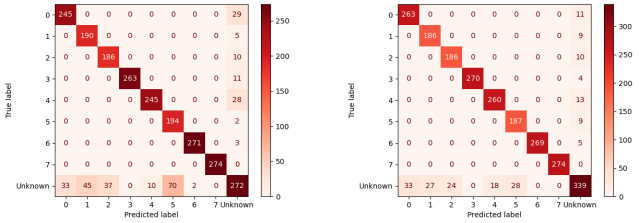

Fig. 6. Confusion matrices of the naive (a) ViT and (b) PLViT. The known classes are labeled as 0–7 and all unknown samples are labeled as "Unknown."

2) **AUIN (or AUOUT)** is the area under the precision-recall curve when setting in-of (or out-of) distribution samples as positive.
3) **Detection accuracy (DTACC)** refers to the maximum classification probability over all possible thresholds.

First, we alternate the civilian classes and military classes of the FGSCR-42 dataset as in/out distribution samples. As shown in Table III, PLViT outperforms other learning frameworks. In addition to this, we also get some other observations. The OOD performances of the CNN-based learning frameworks only based on Euclidean distance metric (GCPL, RPL) are even worse than the baseline method (Softmax), which means that the effectiveness of optimizing the feature representations of fine-grained objects in the distance space is limited. For learning frameworks other than PLViT, competitive results can be achieved by using the original ViT directly compared to the CNN-based frameworks. This demonstrates the innate superiority of the ViT-based framework for the recognition of fine-grained objects. Both the TNR and AUROC of PLViT are much better than the naive ViT with only the overall representation, which proves that PLViT can effectively differentiate fine-grained OOD classes that are similar in overall representation with in-distribution classes.

Then, we select eight and two classes of the MSTAR dataset as known and unknown classes and show the recognition results of the naive ViT and PLViT by plotting the confusion matrix. The selected known and unknown classes are labeled as 0–7 and "Unknown," respectively. As shown in Fig. 6, the unknown class samples are easy to be misclassified as one of the known classes by the naive ViT. However, the proposed PLViT can effectively avoid such misclassification that most unknown samples are correctly recognized while maintaining the classification accuracy of known classes.

TABLE IV
ANALYSIS OF COMPUTATIONAL COMPLEXITY

Methods	Image Size	FLOPs	Params
ViT	$224 \times 224 \times 3$	17.6G	85.8M
TransFG	$224 \times 224 \times 3$	25.6G	85.9M
PLViT (Coarse)	$224 \times 224 \times 3$	16.1G	78.7M
PLViT (Fine)	$224 \times 224 \times 3$	17.6G	86.4M

3) Computational Complexity: Besides the performance comparison based on different metrics, we also provide the analysis of computational complexity (FLOPs) and parameter number for different transformer-based methods in Table IV. First, we can see that the computational complexity of transformer architecture for fine-grained recognition (TransFG) increases significantly, which is mainly due to the generation of overlapping patches. In contrast, both the parameter number and computational complexity of our proposed method in the coarse optimization stage for overall feature representation are lower than ViT, and the incorporation of local feature extraction and fusion feature optimization in the fine optimization stage does not bring much increase.

D. Ablation Studies

1) Effectiveness of Overall Feature Representation and Local Feature Extraction: Overall feature representation and local feature extraction are two critical parts of our proposed method, which consist of three main modules. The distance-based optimization L_D for overall feature representation, the CAPS and the multilayer aggregation (MLA) for local feature extraction. We conduct ablation experiments on FGSCR-42 dataset to prove their effectiveness.

The performances are reported by AUROC and OSCR as shown in Table V. The Plain ViT represents the results of the ViT-B-16 backbone. First, we can see that even without the modules for local feature extraction, improving the representation of overall features does provide an effective improvement in OSR performance. Then, we evaluate the performance when using CAPS and MLA for local feature extraction separately. It can be confirmed that both context information and MLA are effective in improving OSR performance. Finally, we combine

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TABLE V

ABLATION EXPERIMENTS TO PROVE THE EFFECTIVENESS OF OVERALL FEATURE REPRESENTATION AND LOCAL FEATURE EXTRACTION

Methods	L_D	CAPS	MLA	AUROC	OSCR
Plain ViT				91.4	90.3
w/o MLA and CAPS	✓			93.1	92.5
w/o MLA	✓	✓		93.9	93.4
w/o CAPS	✓		✓	94.1	93.5
PLViT	✓	✓	✓	**94.6**	**94.2**

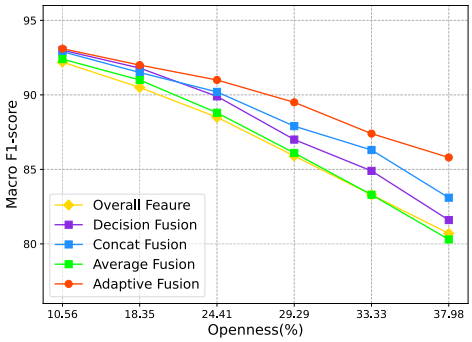


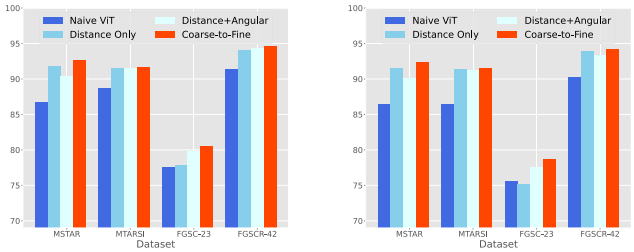
Fig. 7. Macro F1-scores against varying openness with different fusion methods.

them to constitute the proposed PLViT and achieve the best OSR performance.

2) Effectiveness of Adaptive Feature Fusion: We propose an adaptive fusion method to integrate the overall and local features, which will then be optimized in the unified angular space. To prove the effectiveness of the proposed fusion method, we set the ten civilian classes of FGSCR-42 dataset as known classes, and vary the number of unknown military classes from 5 to 32, corresponding to the openness from 10.56% to 37.98%.

We report the performance by macroaveraged F1-score as shown in Fig. 7. The Overall Feature represents the results of thresholding the overall feature-based Softmax scores. The Decision Fusion represents the decision-level fusion method of the global and local feature-based predictions. The Concat Fusion and Average Fusion refer to using the parameterless fusion methods for overall and local features. First, the decision fusion method can improve the overall feature-based performance to some extent, but such improvement is limited. Then, there is no significant improvement in averaging the overall and local features compared with only using the well-represented overall feature. Moreover, concatenating the overall and local features allows for better performance, but there is a tendency for its improvement to decay as the openness increases. Finally, adaptively fusing the overall and global features can lead to the best results, especially when meeting a large openness.

3) Effectiveness of Coarse-to-Fine Optimization Strategy: Following the human perception mode, we propose a coarse-to-fine optimization strategy to progressively learn the decision boundaries for fine-grained objects. In this experiment, we compare the OSR performance of the proposed method trained with different optimization strategies on various datasets. The AUROC and OSCR are shown in



(a) (b)

Fig. 8. Ablation experiments for the coarse-to-fine optimization strategy. (a) AUROC. (b) OSCR.

TABLE VI

OSR PERFORMANCE WITH VARIOUS VARIABILITIES

Methods	None	Illumination	Noise	Both
ViT	88.7	87.0(-2.0%)	86.2(-2.8%)	85.9(-3.2%)
PLViT	91.7	91.2(-0.5%)	90.6(-1.2%)	90.4(-1.4%)

Fig. 8(a) and (b), respectively. The Distance Only refers to the optimization strategy in the single distance space. Different from our proposed coarse-to-fine optimization strategy, the Distance + Angular refers to simultaneously optimizing all modules of the model without the coarse-to-fine sequence. According to the results, we can see that the proposed PLViT trained by different optimization strategies can outperform the naive ViT on most datasets. Among different optimization strategies, our proposed coarse-to-fine optimization strategy can consistently lead to the best OSR performance.

4) Performance When Meeting Variabilities: There exist various variables during the collection of remote sensing images, which will bring more difficulties to the recognition [71]. In this experiment, we evaluate the OSR performance of the proposed method on MTARSI dataset when meeting changes in illumination and interference of noise. Specifically, we simulate the two variabilities above by multiplying the pixel values of each image with randomly-generated factors in [0.5, 1.5] and adding a 10-dB Gaussian noise to the images, respectively. The results are reported by AUROC in Table VI, and degrees of performance degradation with different variabilities are shown in brackets. According to the results, we can see that the performance degradation of the proposed method is smaller than that of the naive ViT, which demonstrates better applicability in real-world applications.

E. Parameter Analysis

We analyze the impact of different settings of the hyperparameters of PLViT, namely, the distance loss weight λ_{Dis}, the context coefficient η, the angular loss weight λ_{Ang}, and the angular margin θ. First, we vary λ_{Dis} from 0.001 to 0.5 when optimizing the overall feature representation and the experimental results are shown in Fig. 9(a). We can see that the OSR performance can be improved by increasing λ_{Dis} in a certain range. However, an overlarge value of λ_{Dis} can degrade the OSR performance. This indicates that the proposed method is sensitive to an overlarge value of λ_{Dis} as it will affect the learning of the classification function.

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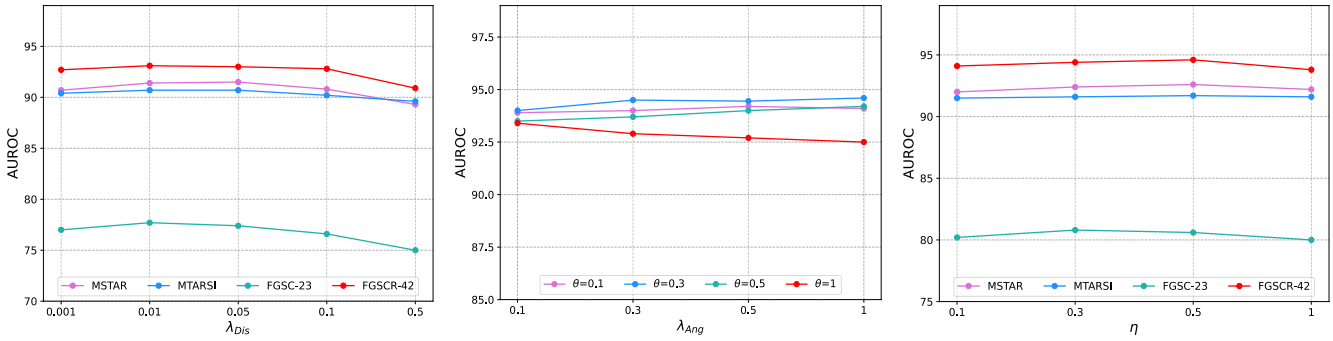

(a) (b) (c)

Fig. 9. OSR performances of the proposed method with different settings of (a) λ_{Dis}, (b) λ_{Ang}, θ, and (c) η.

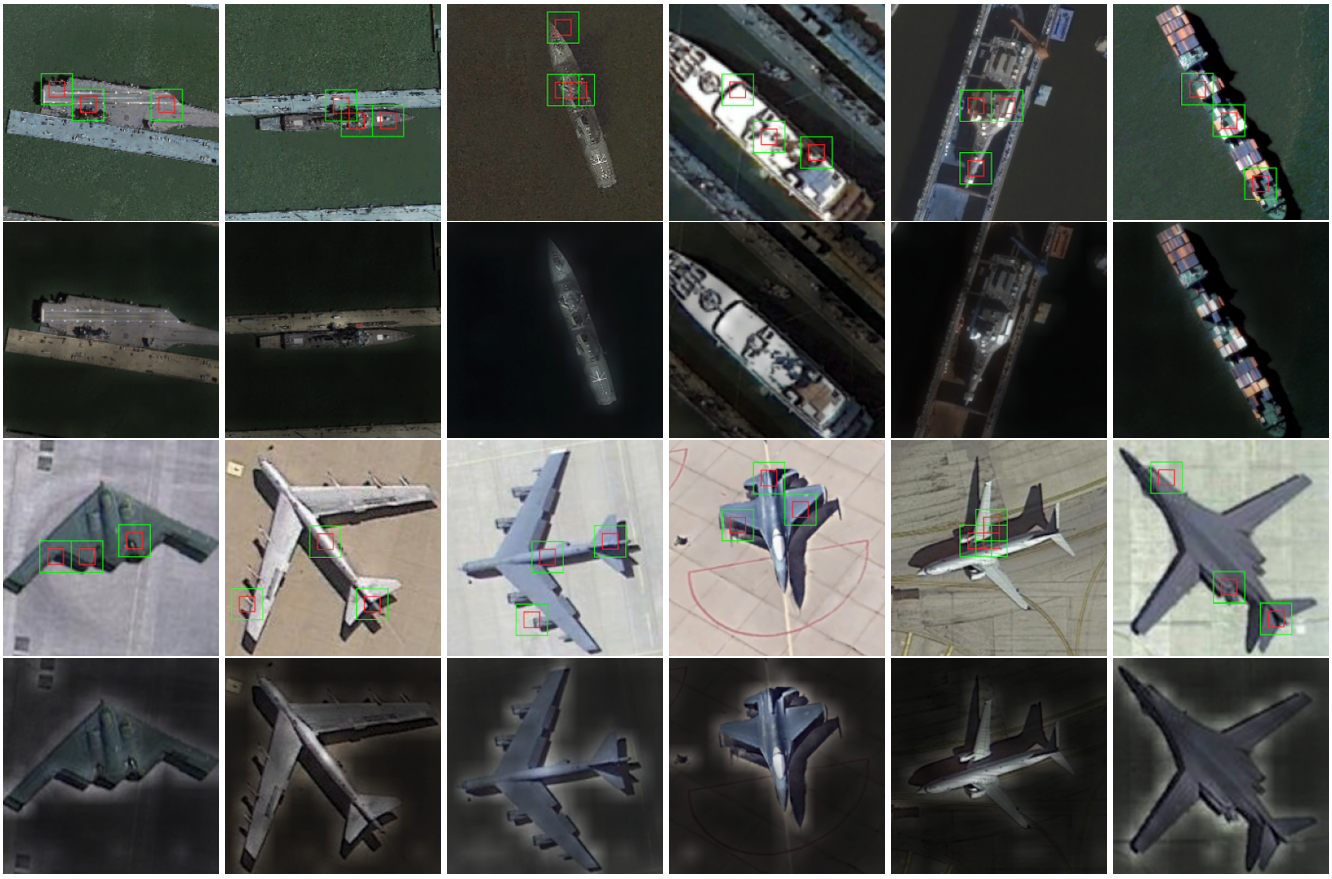

Fig. 10. Visualization results of PLViT on FGSCR-42 and MTARSI datasets. The discriminative part regions can be accurately located and the objects can be successfully separated from the background (best viewed in color).

Then, we fix λ_{Dis} and η to 0.01 and 0.5, and evaluate the OSR performance on the FGSCR-42 dataset with the combinations of different values of λ_{Ang} and θ for fusion feature optimization. According to the results shown in Fig. 9(b), the proposed method is more sensitive to θ than λ_{Ang}, and a moderate value of θ is more appropriate.

Finally, we fix λ_{Dis}, λ_{Ang}. and θ to 0.01, 1, and 0.3, and vary η from 0.1 to 1. As shown in Fig. 9(c), the OSR performance of the proposed method remains stable across a wide range η, but tends to decrease when η is too large. This is because overemphasizing the context information will lead to the loss of local details.

F. Qualitative Analysis

We conduct qualitative analysis of our proposed method to provide a clear explanation of the recognition process. Visualization results on some randomly sampled images from FGSCR-42 and MTARSI datasets are shown in Fig. 10. The first and third rows of Fig. 10 demonstrate the positions of the selected patches and their surrounding regions for local feature extraction. To ensure the clarity of visualization, we only plot the top-three local patches in red [ranked by the attention score calculated in (10)] with the bounding boxes in green that cover the center points of their neighboring patches. The second and fourth rows show the overall attention maps of the whole

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images by averaging the weights of all self-attention heads. The results indicate that our proposed method can effectively capture the discriminative part regions for fine-grained objects in remote sensing images. Moreover, according to the overall attention maps, our proposed method can precisely separate the target objects from the complex background in remote sensing images.

V. Conclusion

In this article, we have proposed a PLViT for OSR of fine-grained objects in remote sensing images. The proposed PLViT overcomes the limitations of existing OSR methods on fine-grained datasets, and it can learn more discriminative decision boundaries by leveraging both the overall and local features in a coarse-to-fine manner. We propose a context-aware patch selection module to locate discriminative part regions, and the aggregated multilayer representations can provide a more comprehensive description of subtle details for local feature extraction. Moreover, the inconsistency between the scales of fusion features and overall-based memories is effectively avoided by the optimization strategy in the angular space. We have validated the effectiveness of the proposed PLViT by comparing it with other state-of-the-art methods on various benchmarks, and the experimental results show that PLViT can achieve superior performance. Qualitative analysis is also conducted to prove the interpretability of our method. In the future, we will try to further explore the open-world deep learning methods in remote sensing areas, such as open-set detection and open-set incremental learning.

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